Agentic AI: Capabilities, Architectures and Use Cases

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Executive summary

Agentic AI is AI that pursues a goal on its own: it plans steps, uses tools, checks results and adjusts, instead of answering one prompt and stopping. Traditional AI predicts, classifies or generates when asked; an agent decides what to do next.

Every agent runs the same loop of four capabilities: perceive, reason, act and learn. One agent running that loop suits focused tasks. Several specialised agents working together suit large, multi-step work, at the cost of more coordination.

This report explains each idea and gives concrete use cases for each, then covers the main risks.

What is Agentic AI?

Agentic AI is a system that is given a goal and works toward it autonomously over many steps. It usually has a large language model (LLM) at its core, connected to tools such as web search, code execution, databases, APIs and email. It keeps memory of what it has done and decides its next action from the result of the last one.

Traditional AI is reactive: a spam filter labels an email, a recommender ranks products, a chatbot answers a question. Each output is one step, and a human or a fixed program decides what happens next. Agentic AI moves that decision into the system.

Example. Asked "book me a trip to Delhi next week under ₹20,000", a traditional assistant lists flight options. An agent searches flights and hotels, compares prices against the budget, checks your calendar, books, and emails you the itinerary.

Aspect	Traditional AI	Agentic AI
Unit of work	One task: predict, classify, generate	A goal broken into many steps
Who decides the next step	A human or a fixed program	The agent itself

Aspect	Traditional AI	Agentic AI
Interaction	Single prompt → single response	Continuous loop until the goal is met
Tools	None, or fixed integrations	Chooses and calls tools as needed
Memory	Usually stateless	Short-term and long-term memory
Adapting to change	Retrained offline	Adjusts its plan during the task
Human role	Operator at each step	Sets the goal, supervises, approves key actions
Examples	Spam filter, image classifier, basic chatbot	Coding agent, research agent, autonomous support agent

Core capabilities: Perceive, Reason, Act, Learn

Every AI agent works through a loop of four capabilities, repeated until the goal is met. Each pass uses what the last action produced, which is what lets an agent recover from errors mid-task.



the agent loop · 4 capabilities

In a support agent, for example: it reads a complaint (perceive), decides a refund is due under policy (reason), issues the refund and replies (act), then records the case so similar tickets are handled faster (learn).

Single-agent vs multi-agent systems

A single-agent system is one agent with one set of tools handling the whole task. A multi-agent system splits the work across several agents, each with its own role, tools and instructions, that pass work between them.

Multi-agent systems are usually organised in one of three patterns:

- **Orchestrator–worker (hierarchical):** a lead agent breaks the goal into subtasks, hands them to worker agents, and combines their results.
- **Sequential pipeline:** agents work in a fixed order, each passing its output to the next, like researcher → writer → editor.
- **Peer collaboration or debate:** agents with different views critique or vote on each other's output to reach a better answer.

Aspect	Single-agent	Multi-agent
Structure	One agent, one loop	Several specialised agents
Best for	Focused, well-defined tasks	Large, complex or parallel tasks
Strengths	Simple to build, debug and audit; low cost	Specialisation, parallel work, cross-checking
Weaknesses	Context overload on big tasks; one point of failure	Coordination overhead, higher cost, harder to trace errors
Speed	Steps run one after another	Subtasks can run in parallel
Example	A personal email-triage agent	A software team of planner, coder, tester and reviewer agents

A practical rule: start with one agent, and add agents only when a task clearly separates into roles or needs parallel work.

Use cases

The tables below give use cases for each topic in this report: agentic AI in general, each core capability, and each system type.

Agentic AI across industries

Industry	Use case	What the agent does on its own
Customer service	Autonomous support agent	Reads the ticket, checks order status, issues a refund within policy, replies, escalates edge cases
Software development	Coding agent	Reads a bug report, finds the code, writes a fix, runs tests, opens a pull request
Finance	Fraud and compliance agent	Flags unusual transactions, gathers evidence, freezes accounts or files reports for review
Healthcare	Clinical admin assistant	Summarises patient records, drafts notes, schedules follow-ups, handles insurance prior authorisation
E-commerce	Shopping assistant	Finds products to a budget, compares sellers, applies coupons, places the order
Supply chain	Inventory agent	Forecasts demand, reorders stock, reroutes shipments when a supplier is delayed
Research	Research agent	Searches papers and the web, reads sources, writes a cited summary
IT operations	Incident response agent	Detects an outage, checks logs, restarts services, writes the incident report

Use cases for each core capability

Capability	Use case example
Perceive	A self-driving car reads cameras and lidar; a support agent reads an incoming email and its attachments
Reason	A travel agent plans a trip within budget and dates; a coding agent works out which file causes a bug

Capability	Use case example
Act	An agent books the flight, sends the email, updates the CRM or runs the code
Learn	A recommendation agent improves from user feedback; a support agent stores resolved cases to answer faster next time

Use cases for single-agent and multi-agent systems

System type	Use case	Why this type fits
Single-agent	Personal assistant managing email and calendar	One user, one clear domain, few tools
Single-agent	FAQ and order-tracking chatbot	Narrow, repeatable task
Single-agent	Code completion and bug-fix agent in an IDE	Focused task inside one codebase
Multi-agent	Software team: planner, coder, tester, reviewer	Distinct roles that check each other
Multi-agent	Deep research: lead agent plus parallel search agents	Many sources searched at once, then merged
Multi-agent	Hospital operations: bed, staffing and pharmacy agents	Separate departments that must coordinate
Multi-agent	Warehouse robots coordinating routes	Many physical agents sharing one space
Multi-agent	Content pipeline: researcher → writer → editor → SEO	Each step needs different skills

Challenges and conclusion

Autonomy brings new risks, because an agent's mistakes become actions, not just wrong answers.

- **Reliability:** small errors compound over many steps, and LLMs can hallucinate facts or tool inputs.
- **Safety and security:** an agent with access to email, money or code can do real damage; prompt injection in web pages or documents can hijack it.
- **Accountability:** it can be unclear who is responsible when an agent acts wrongly.
- **Cost and speed:** many model calls per task, especially in multi-agent systems, raise cost and latency.
- **Evaluation:** multi-step behaviour is harder to test than a single prediction.

Common safeguards are human approval for high-impact actions, limited tool permissions, logging every step, and testing agents in sandboxes before real use.

Conclusion. Agentic AI shifts AI from a tool that answers to a system that acts. The perceive–reason–act–learn loop is its foundation; single agents handle focused tasks well, and multi-agent systems scale to complex work. The best results come from matching the design to the task and keeping humans in control of decisions that matter.